Homework 23: 9.2-9.6 **Name** _____

AP Calculus BC **Due** _____

Period _____ **Score** _____ **/ 100**

Skill Builder 9.2 – Second Derivatives of Parametric Equations

In the following exercises, find $\frac{d^2y}{dx^2}$ and determine the intervals for t in which the curve is concave up or concave down.

1.) $x = t^2 + 1,\ y = t^2 + t$	**2.)** $x = e^t,\ y = te^{-t}$
3.) $x = 2t - \ln t,\ y = 2t + \ln t$	**4.)** $x = \sin t,\ y = 2\cos t \quad 0 \le t \le 2\pi$

Skill Builder 9.3 – Finding Arc Lengths of Curves Defined by Parametric Equations

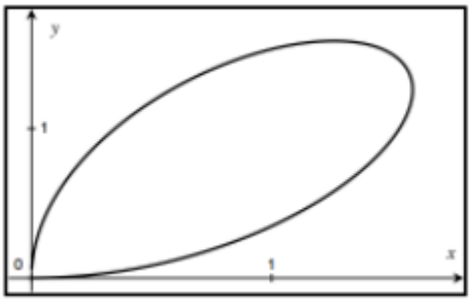
Find the arc length of the given curve on the indicated interval. Calculators are permitted only for 3 and 4.

1.) $x=\sin^3 t$ and $y=\cos^3 t$ $\left[0,\frac{\pi}{2}\right]$	**2.)** $x=2t^{3/2}$ and $y=1+(8-t)^{3/2}$ $[0,4]$
3.) $x=t^2-t$ and $y=4t^3+2$ $[-1,1]$	**4.)** $x=2t\sin t$ and $y=3\cos t$ $[0,2\pi]$
5.) $x=\sqrt{2}e^{-t}\sin t$ and $y=\sqrt{2}e^{-t}\cos t$ $[0,\pi]$	
6.) $x=\frac{t^5}{10}+\frac{1}{6t^3}$ and $y=t$ $[1,3]$	

7.) The path of a soccer ball is modeled by the equations $x = 80t\cos 30^\circ$ and $y = 80t\sin 30^\circ - 16t^2$ where x and y are measured in feet. Graph the path of the ball and find the difference between the horizontal distance the ball travels and the arc length of the path of the ball. Be sure to label your axes.

8.) Given the parametric equations

$x = \dfrac{3t}{1+t^3}$ and $y = \dfrac{3t^2}{1+t^3}$, $t \geq 0$, sketch it on your calculator.

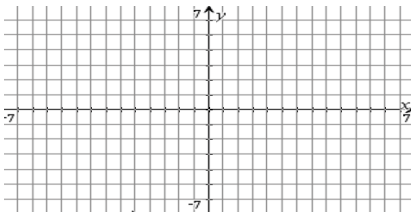
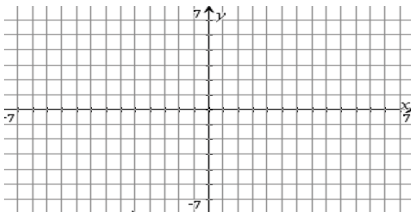
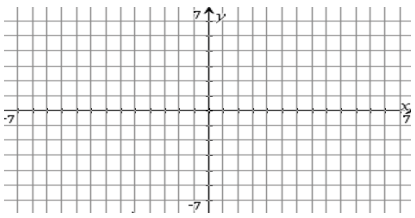


a) Other than at $t = 0$, the graph has one horizontal tangent and a vertical tangent. Determine the difference in the t values at these points.

b) Approximate the arc length of the closed loop.

Skill Builder 9.4: Defining and Differentiating Vector-Valued Functions

For each of the following, write the vector in component form and sketch it in its component form.

1. initial point: $(1,3)$ terminal point: $(5,6)$ 	**2.** initial point: $(4,-3)$ terminal point: $(-2,-4)$ 	**3.** initial point: $\left(\frac{3}{2},-\frac{1}{2}\right)$ terminal point: $\left(-4,\frac{14}{3}\right)$ 

For each of the following, find the vector **v** where **u** = $\langle 5,-3\rangle$ and **w** = $\langle 1,4\rangle$. Write in component form.

4. $\mathbf{v}=\frac{3}{4}\mathbf{u}$	**5.** $\mathbf{v}=2\mathbf{w}-\mathbf{u}$	**6.** $\mathbf{v}=\frac{1}{2}(-5\mathbf{u}-3\mathbf{w})$

For each of the following, find *a* and *b* such that $\mathbf{v}=a\mathbf{u}+b\mathbf{w}$ where **u** = $\langle 1,2\rangle$ and **w** = $\langle 1,-1\rangle$.

7. $\mathbf{v}=\langle 2,1\rangle$	**8.** $\mathbf{v}=\langle 0,3\rangle$	**9.** $\mathbf{v}=\langle -1,7\rangle$

Given that $\mathbf{u} = \langle 2,-3 \rangle$ and $\mathbf{v} = \langle 4,4 \rangle$, find each of the following.

10. $\|\mathbf{u}\|$	**11.** $\|\mathbf{v}\|$	**12.** $\|\mathbf{u}+\mathbf{v}\|$
13. $\left\|\dfrac{\mathbf{u}}{\|\mathbf{u}\|}\right\|$	**14.** $\left\|\dfrac{\mathbf{v}}{\|\mathbf{v}\|}\right\|$	**15.** $\left\|\dfrac{\mathbf{u}+\mathbf{v}}{\|\mathbf{u}+\mathbf{v}\|}\right\|$
16. $\mathbf{u}\cdot\mathbf{v}$	**17.** The angle between $\mathbf{u}$ and $\mathbf{v}$	

For each of the following, find a unit vector **(a)** parallel and **(b)** orthogonal to the graph of $f(x)$ at the given point.

18. $f(x) = x^3 \quad (1,1)$	**19.** $f(x) = \sqrt{25-x^2} \quad (3,4)$

For each of the following, determine whether **u** and **v** are orthogonal, parallel or neither.

20. $\mathbf{u} = 4\mathbf{i} - 6\mathbf{j}, \quad \mathbf{v} = -2\mathbf{i} + 3\mathbf{j}$	**21.** $\mathbf{u} = \langle 2,18 \rangle, \quad \mathbf{v} = \left\langle \frac{3}{2}, -\frac{1}{6} \right\rangle$	**22.** $\mathbf{u} = \langle \tan\theta, -1 \rangle, \quad \mathbf{v} = \langle \cos\theta, \sin\theta \rangle$

For each of the following, evaluate the limit.

26. $\lim\limits_{t\to 3}\left(\frac{1}{t}\mathbf{i}+\frac{t^2-9}{t-3}\mathbf{j}\right)$	**27.** $\lim\limits_{t\to 1}\left(\sqrt[3]{t}\,\mathbf{i}+\frac{\ln t}{t^2-1}\mathbf{j}\right)$	**28.** $\lim\limits_{t\to\infty}\left(4e^{-2t}\mathbf{i}+\frac{5}{t^2-1}\mathbf{j}\right)$

For each of the following, determine the intervals where the vector-valued function is continuous.

29. $\mathbf{v}(t)=\sqrt{t}\,\mathbf{i}+\sqrt{t-2}\,\mathbf{j}$	**30.** $\mathbf{v}(t)=\sin(t)\,\mathbf{i}-\cos(t)\,\mathbf{j}$	**31.** $\mathbf{v}(t)=e^{-t}\,\mathbf{i}+\sin^{-1}(t)\,\mathbf{j}$

Skill Builder 9.5: Integrating Vector-Valued Functions

For each of the following, evaluate the integral.

1. $\int\left[(3t-1)\mathbf{i}-4t^3\mathbf{j}\right]dt$	**2.** $\int\left[\sec^2(t)\mathbf{i}-\frac{1}{\sqrt{1-t^2}}\mathbf{j}\right]dt$	**3.** $\int_0^{\pi/2}\left[(\cos 2t)\mathbf{i}+(\sin 3t)\mathbf{j}\right]dt$

For each of the following, find $\mathbf{r}(t)$ using the given conditions.

4. $\mathbf{r}'(t) = \left(4e^{2t}\right)\mathbf{i} + \left(3e^{t}\right)\mathbf{j},\ \mathbf{r}(0) = 2\mathbf{i}$	**5.** $\mathbf{r}''(t) = -4\cos(t)\mathbf{i} - 6\mathbf{j},$ $\mathbf{r}'(0) = 3\mathbf{j},\ \ \mathbf{r}(0) = 2\mathbf{i}$

Find the vector valued function $\mathbf{f}(t)$ that satisfies the given initial condition(s).

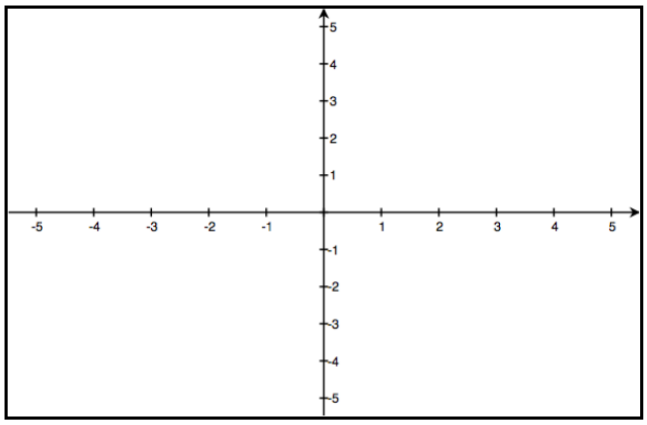
6. $\mathbf{f}(0) = \left\langle \frac{1}{2}, -1 \right\rangle,\ \mathbf{f}'(t) = \left\langle te^{-t^2}, -e^{-t} \right\rangle$
7. $\mathbf{f}'(0) = \langle 3, 0 \rangle,\ \mathbf{f}(0) = \langle 0, 3 \rangle,\ \mathbf{f}''(t) = \langle 5\cos t, -2\sin t \rangle$

8. $\mathbf{f}'(0)=\langle 0,2\rangle$, $\mathbf{f}(0)=\langle 3,0\rangle$, $\mathbf{f}''(t)=\langle 4t^3,3t^2\rangle$

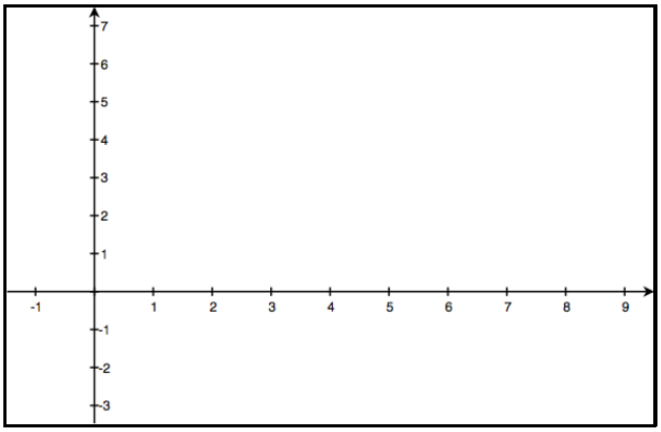
Skill Builder 9.6: Solving Motion Problems Using Parametric & Vector-Valued Functions

For each of the following problems, $\mathbf{r}$ describes the path of an object in the *xy* plane. Generate the velocity and acceleration vectors and the speed at the given point. Sketch the path, velocity and acceleration vectors at that point using two different colors.

1. $\mathbf{r}(t)=(5-t)\mathbf{i}+(t-1)\mathbf{j}$ at $t=2$

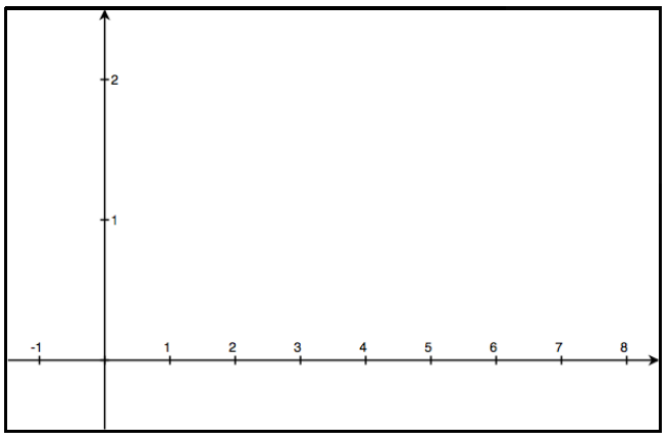


2. $\mathbf{r}(t)=(8-t^2)\mathbf{i}+(3t-3)\mathbf{j}$ at $t=0$

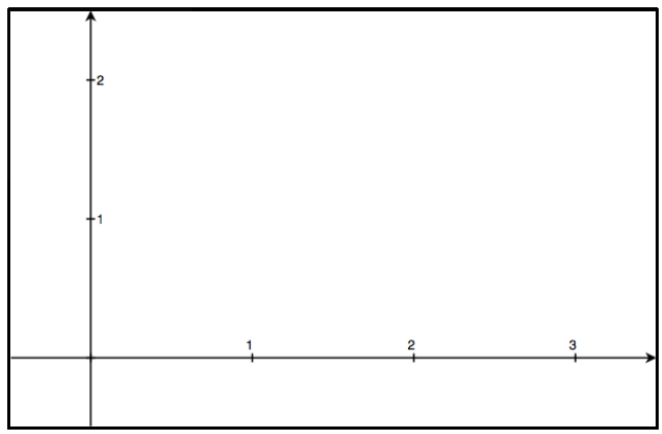


You may use your graphing calculator to sketch the motion path for Problems 3 and 4.

3. $\mathbf{r}(t) = (t - \sin t)\mathbf{i} + (1 - \cos t)\mathbf{j}$ at $t = \pi$

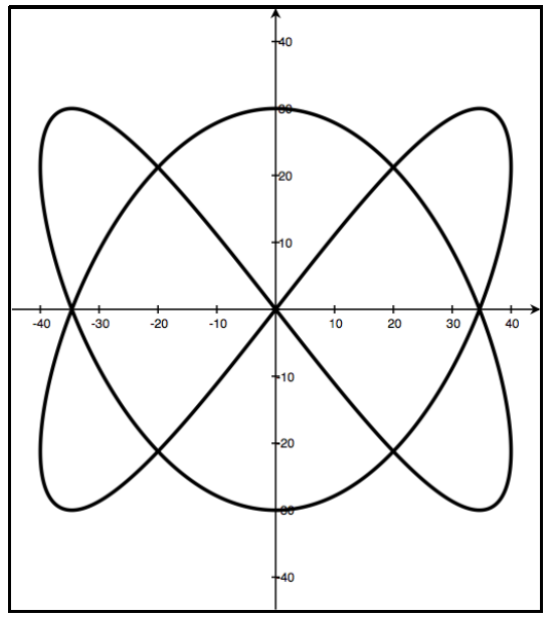


4. $\mathbf{r}(t) = \left(e^{t}\right)\mathbf{i} + \left(e^{-t}\right)\mathbf{j}$ at $t = 0$



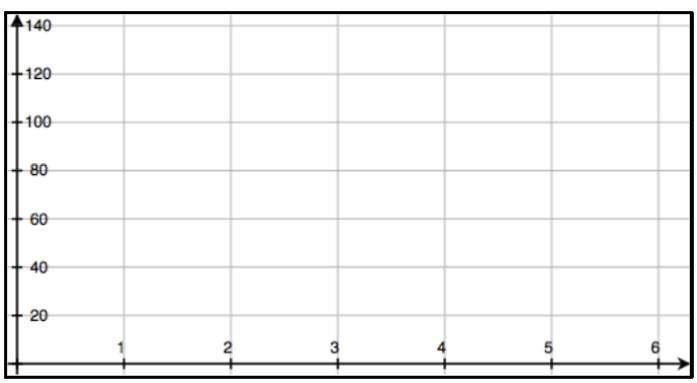
5. Use the acceleration function $\mathbf{a}(t) = e^{-t}\mathbf{i} + \sin(t)\mathbf{j}$ to find the velocity and position function of an object at rest whose initial position is $(0,0)$ at $t = \pi$.

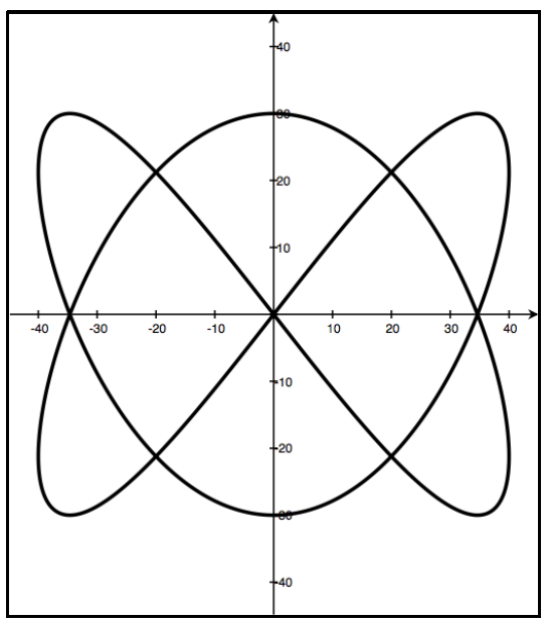
6. "The Jumble," an amusement park ride, has people riding through the air as shown in the graph to the right.
The path described with the vector-valued function $\mathbf{r}(t) = (40\sin 2t)\mathbf{i} + (30\cos 3t)\mathbf{j}$.



a.) Find both the velocity and acceleration vectors at **i.)** $t = \frac{3\pi}{4}$ and **ii.)** $t = \frac{7\pi}{3}$. Sketch the velocity vectors.

b.) Find and sketch the speed of a rider on "The Jumble" on the time interval $0 \le t \le 2\pi$. Find the values of t where the speed has relative extrema and absolute extrema and place their locations on the path diagram below. Explain why these locations make sense.





Enrichment: Unit 9 – Parametric and Vector-Valued Functions (Circuit)

Begin in the first cell marked #1 and find the solution of the given problem. To advance in the circuit, search for your answer and mark that cell #2. Continue in this manner until you complete the circuit. Show all pertinent work. Note: Most of the problems will require the use of your graphing calculator.

# 1 **Ans:** 0.331 Given $x = 2t^3 + 5,\ y = 6t^5 - 12t^4 + 7,$ find $\dfrac{dy}{dx}$.	# ___ **Ans:** $y - 5 = 0.902(x+1)$ A particle moves along a curve so that the velocity vector is given by $\mathbf{v}(t) = \left\langle 3\sin(e^t), 5^{\ln t} \right\rangle,\ t > 0.$ Find the acceleration vector at *t* = 2.
# ___ **Ans:** 4 A particle moves along a curve so that $\dfrac{dx}{dt} = e^{t/3}$ and $\dfrac{dy}{dt} = \cos(5t)$. What is the speed of the particle when *t* = 2?	# ___ **Ans:** 6.363 An object moving along a curve in the *xy*-plane has $\dfrac{dy}{dt} = 3 + e^{\sin t}$. The derivative $\dfrac{dx}{dt}$ is not explicitly given. At time *t* = 2, the value of $\dfrac{dy}{dx}$ is -5.4. Find the value of $\dfrac{dx}{dt}$ when *t* = 2.

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Ans: 0.647

A particle moves along a curve so that $\frac{dx}{dt} = e^{\cos t}$ and $\frac{dy}{dt} = t^2 + 3\sin t$. At *t* = 1, the particle is at the point $(4, -7)$. Find the position of the particle a the point when *t* = 2.

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Ans: $\langle 3.979, -15.832 \rangle$

A particle moves in the *xy*-plane so that $\frac{dx}{dt} = 3t - \cos(t^2)$ and $\frac{dy}{dt} = 4t + \sin t$. For $0 \le t \le 1$, what is the value of *t* at which the line tangent to the path of the particle is vertical?

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Ans: 2.121

A particle moves in the *xy*-plane so that $\frac{dx}{dt} = \sin\left(t^2\right)$ and $\frac{dy}{dt} = e^{\cos t}$. At *t* = 3, the particle is at the position $(-1, 5)$. Write the equation of the tangent line to the curve at the point where *t* = 3.

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Ans: -1.015

A particle moves along a curve in the *xy*-plane so that its position is given by $x(t) = 3t + \sin\left(t^2\right)$ and $y(t) = 5 + \cos\left(e^{2t}\right)$, $0 \le t \le \pi$. Find the velocity vector when the particle's horizontal position is *x* = 4.

# ___ **Ans:** $5t^2 - 8t$ Use your answer to the previous problem to find $\frac{d^2y}{dx^2}$.	# ___ **Ans:** $(5.113, -1.797)$ A particle moves along a curve so that $\frac{dx}{dt} = \sin\left(e^{1/4}\right)$ and $\frac{dy}{dt} = \sqrt{t^2+5}$. Find the total distance traveled by the particle over the time interval $1 \le t \le 3$.
# ___ **Ans:** $\langle 9.939, 2.455 \rangle$ A particle moves along a curve so that $\frac{dx}{dt} = \cos\left(t^2\right)$ and $\frac{dy}{dt} = e^{\sin t}$. For what value of *t*, $0 < t < 1$, does the tangent line to the curve have a slope of 2?	# ___ **Ans:** $\frac{10t-8}{6t^2}$ The position of a particle in the *xy*-plane is given by the parametric equations $x(t) = t^3 - 6t^2$ and $y(t) = 2t^3 - 9t^2 - 24t$. For what value of *t* is the particle at rest?